

Nuts and Bolts of New Ventures

Joseph Hadzima

MIT 15.393 OpenCourseWare entrepreneurship course
Lecture notes organized by [LazyingArt LLC](#) with [Video2Book](#)

Original lectures by Joseph Hadzima for MIT OpenCourseWare. Lecture notes organized by [LazyingArt LLC](#).

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Chapter 1

Introduction: Most Startups Fail; How to Improve Your Odds

Overview. Joseph Hadzima begins this MIT OpenCourseWare course the right way: not with a doctrine, not with a slogan, but with a room full of motives and questions. He first asks why we are here, then what we do not know, then why we would attempt something so painful given such poor odds. Only after that does he permit himself equations, frameworks, and diagrams. These notes follow that sequence closely. They are curated by LazyingArt LLC from the source lecture, and they are meant to read as companion notes for a practical course in venture formation rather than as startup mythology.

1.1 Why We Are Here

The lecture opens with a guide's promise. We are beginning a six-session journey through the nuts and bolts of new ventures, and the first task is to identify the motives that have brought the room together. Hadzima does not sort those motives into pure and impure kinds. He simply lays them out and lets us hear ourselves in them.

Some are here because they have never taken a formal class in entrepreneurship and want to see what the subject even looks like. Some are here because they have encountered some miserable product or process and want to make it less bad. Some have an invention and want to bring it to life. Some want to change the world. Some are drawn by the glamour and money that entrepreneurship still advertises.

The lecture's early tone matters. We are not yet being asked to judge a venture. We are being asked to notice why the possibility of a venture exerts a pull at all.

The Dropbox anecdote gives this opening its first concrete shape. A student, stuck without access to his files while traveling, does not merely complain. He converts irritation into a design imperative: make a storage system that does not "suck." The lecture uses this story carefully. It is not yet a theory of ventures. It is an example of how dissatisfaction, if pursued, can become the beginning of disciplined entrepreneurial thought.

1.2 The Entrepreneur's First Questions

From motive, the lecture widens into uncertainty. Hadzima says that he will give us guideposts and asks us to carry his questions from session to session. The effect is cumulative. We do not begin with a neat framework because the subject does not first arrive to us neatly.

1. How do we start?
2. What, exactly, do we do first?
3. What problem are we solving?
4. Does the proposed solution really answer that problem?
5. Who cares about the idea, specifically, and who is the customer?
6. How will the venture make money?
7. How long will it take to get to market, what will it cost, and what resources will be required?
8. What legal entity, if any, should be formed?
9. How can the idea be protected, if protection is the right objective?
10. Will cofounders be needed, and how will the relationship be structured?
11. How will negotiation enter, with employees, consultants, customers, advisors, and investors?
12. How do we discover what we do not know?

The lecture then adds two important qualifications. First, entrepreneurs are always negotiating because they are assembling people and resources they do not yet fully control in order to pursue a vision they do not yet fully own. Second, mistakes are not the scandal. Repeating the same mistake is the scandal. The people teaching the course, he says, have scars, and part of their use is that they may help us recognize recurring potholes before we drive directly into them.

By this point the chapter has already adopted the lecture's rhythm. We begin from personal motives, but we are not left in autobiography. We are pushed outward into operational questions. The subject becomes practical before it becomes formal.

1.3 Why Do This If Most Startups Fail?

At exactly the point where the lecture might have become merely energizing, Hadzima turns and creates resistance. Why do this at all? Why enter a line of work defined by difficulty, embarrassment, and attrition?

The success image is easy to summon: fame, fortune, public recognition. The lecture then opposes to that image the much harsher testimony of startup life. The Nvidia anecdote serves this purpose. The founder, looking back across decades, says in effect that if he had fully known what the road required, he might not have taken it. The point is not anti-entrepreneurial. It is corrective. We are meant to feel the cost before the course offers any technique for improving our odds.

The lecture gives the base rates in blunt numerical form:

$$\Pr(\text{startup success}) \approx 0.10 \approx 10\%, \quad (1.1)$$

$$\Pr(\text{startup failure}) \approx 0.90 \approx 90\%, \quad (1.2)$$

$$\Pr(\text{top-tier VC success per investment}) \approx \frac{3}{10} = 0.300. \quad (1.3)$$

The baseball analogy is part of the argument, not a decorative aside. A venture capitalist succeeding three times out of ten is likened to a .300 hitter: someone who fails often and is still a star. The analogy has a pedagogical use. It takes a discouraging ratio and translates it into a more familiar domain where high failure and excellence already coexist.

One slide, cited in the lecture, sharpens the point further by attributing most failure to poor planning and poor experience. We should not over-read the exact number attached to that slide, but the argumentative purpose is clear. Hadzima is preparing the answer to the question he has just raised. If failure were purely random, there would be little reason for a course like this. If planning and preparation matter, then a course may genuinely improve the odds.

That is why he then introduces the local MIT counterweight. The Kauffman-style numbers are not given as a proof that ventures succeed here routinely; they are given as evidence that when ventures launched in this environment do succeed, they may succeed at very large scale:

$$\frac{26,000}{120,000} \approx 0.217 \approx \frac{1}{5}. \quad (1.4)$$

On the lecture's telling, that means roughly one active MIT-founded company for every five living alumni in the period covered by the report. The same framing attributes annual revenues on the order of \$2 trillion to those companies and says that, if aggregated, they would amount to roughly the eleventh largest economy in the world.

So the lecture does not deny the ugliness of the odds. It accepts them. Then it narrows the ambition. The course will not make venture formation safe. It will try to improve the probability of landing on the successful side of an intrinsically asymmetric game.

1.4 Why This Course, and Who Is In The Room?

After the odds have been stated baldly, the lecture repairs confidence by explaining what kind of course this is. It is not about theory. It is about doing. The speakers are not academics who study entrepreneurship from a distance; they are people who either are doing the work or have already done it.

The course's origin story reinforces the same point. Students effectively demanded a January course on how to start a company. Hadzima, short on time, assembled practitioners and asked them for the most useful possible format: tell the room the three or four or five things you wish somebody had told you when you started. That origin story is itself a small entrepreneurial parable. Demand appeared first; the product was assembled in response.

Then the lecture narrows from the course as a whole to the room immediately in front of it. This matters because the first equation in the lecture does not appear on a pure mathematics slide. It appears on a slide about audience heterogeneity: students and non-student participants, different backgrounds, different interest groups, different expectations.

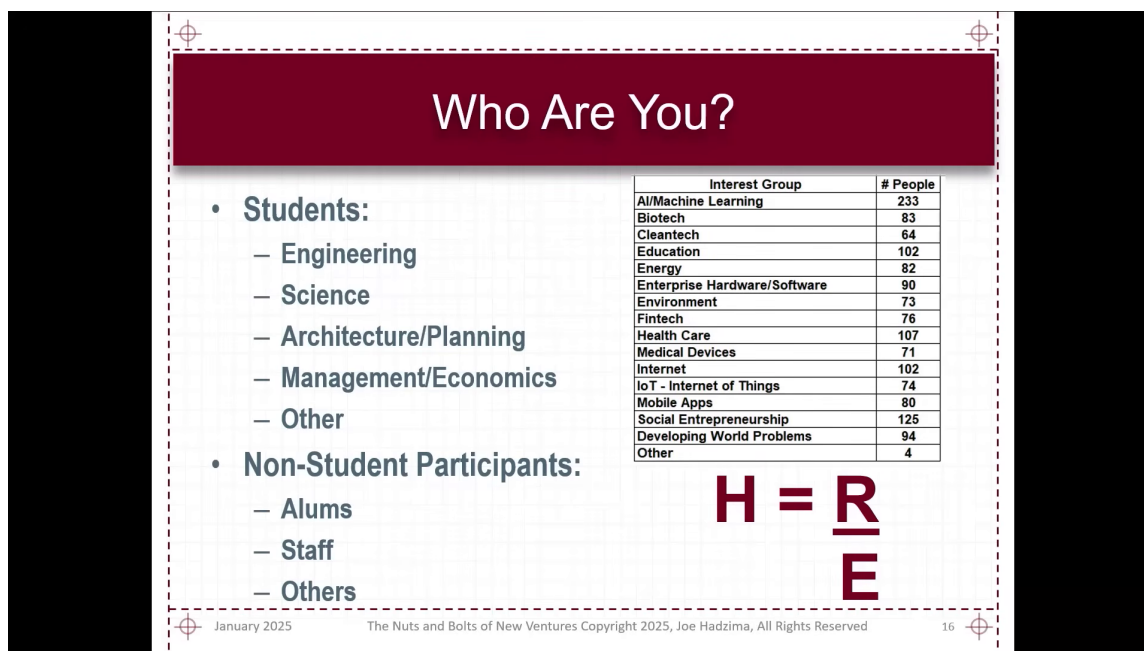


Figure 1.1: Audience mix and the course happiness equation.

The slide is really a room map. The audience breakdown at left and the small interest-group table at right explain why the equation belongs here. The lecture needs a compact rule for thinking about how one course can satisfy people who arrive with different prior knowledge and different hopes.

The slide writes the quotient in a stylized stacked form. We standardize it in ordinary mathematical notation:

$$H = \frac{R}{E}. \quad (1.5)$$

Definition 1.1 (The Happiness Equation). In the lecture's notation,

$$H = \frac{R}{E},$$

where H denotes happiness, R denotes reality, and E denotes expectations.

Hadzima then states the course's objective in deliberately mathematical shorthand. The goal, he says, is to maximize H . We should hear that as lecture rhetoric rather than as a literal optimization problem, but the rhetoric is useful because it forces us to ask what in the ratio can be responsibly changed.

1.4.1 Question & Answer

Question. What are R and E in $H = \frac{R}{E}$?

Answer. R is reality and E is expectations. The lecture briefly records a revealing false start: some Sloan students suggest that R must be revenue and E must be expenses. Hadzima accepts

that reading as a subset and then enlarges it. The equation is not an accounting identity. It is a more general rule for disappointment and satisfaction.

Once we write it cleanly, the lecture’s verbal logic becomes immediate:

$$H = \frac{R}{E}, \quad (1.6)$$

$$H > 1 \iff R > E, \quad (1.7)$$

$$H = 1 \iff R = E, \quad (1.8)$$

$$H < 1 \iff R < E. \quad (1.9)$$

So if we underpromise and overdeliver, then $R > E$, and the ratio exceeds one. If we overpromise and underdeliver, then $R < E$, and the ratio falls below one. The lecture’s little classroom equation is therefore already doing more than managing expectations for a single course. It is establishing a general rule about promises, delivery, and emotional aftermath.

Remark 1.2. The lecture jokes that if we could drive expectations to zero, we might produce infinite happiness. We should not literalize the joke into a usable model. The mathematical content we actually need is comparative: if expectations rise while reality does not, the quotient worsens.

This is also why the equation returns at the end of the lecture, after the discussion of pitch decks and Theranos. It is not a throwaway line. It is one of the lecture’s organizing devices.

1.5 Frameworks for Judging a Venture

Now that the room has been calibrated, Hadzima begins compressing entrepreneurship into reusable filters. The first filter is almost embarrassingly simple, and that is part of its strength.

A venture must create value. If it creates no value, it has no subject matter. But creation alone is not enough. It must also capture enough of that value to continue. In the for-profit setting, that capture eventually takes the form of profit and cash flow. In the not-for-profit setting, it may take the form of enough attention, support, and incoming resources to keep operating. The structural distinction is important. Hadzima is careful not to reduce entrepreneurship to one legal or financial form.

The next compression is the “three whys,” followed by a fourth question that appears only after the first three have begun to work:

1. **Why this?**
2. **Why now?**
3. **Why this team?**
4. **Why won’t this work?**

Why this? The lecture’s strongest illustration is Amy Smith’s water-incubator pitch. Its power begins with scale: 1.9 billion people lacking access to clean water. Only after that scale has been felt does the mechanism appear. Biological contamination testing requires incubation. Existing incubators require electricity. The affected population frequently lacks electricity. Therefore a non-electric incubator attacks a real bottleneck. The structure is exact: large problem, narrow obstacle, specific bridge.

Why now? The FAA example gives this question its form. A technical solution may be genuinely good and still not be timely. But if institutional demand appears at the same moment that the solution becomes available, then timing and opportunity converge. The lecture's point is severe: five years early and five years late may both be failures even when the idea itself is sound.

Why this team? Prior experience can help, but it is not magic. Hadzima's Encore Computer example makes that clear. A team may have astonishing prior credentials and still fail. Prior experience can attract funding; it does not guarantee strategic coherence or executional success. For early founders without that prior experience, the question becomes sharper: what model, what judgment, what surrounding talent gives us reason to believe that this team is the right team?

Why won't this work? This question does not replace the first three. It arrives after them. Hadzima even frames it in sales language: by the time a prospect begins seriously probing limitations, interest already exists. That is why this fourth question belongs to de-risking. We are now asking not whether the venture sounds attractive, but whether its failure modes are understood well enough that we can plan around them.

By this point the lecture has moved from personal motive to evaluative machinery. But it has not yet become abstract. Each filter is anchored in a story, and each story exists to sharpen the filter rather than to replace it.

1.6 Ideas, Execution, Timing, and People

Hadzima says he dislikes frameworks, and then, sensibly, gives us one. The next compression is the lecture's central four-part model of venture success: ideas, execution, timing, and people. The lecture does something subtle here. It first presents the four-part structure in a general, almost static form, and only afterward begins to make one part of it dynamic.

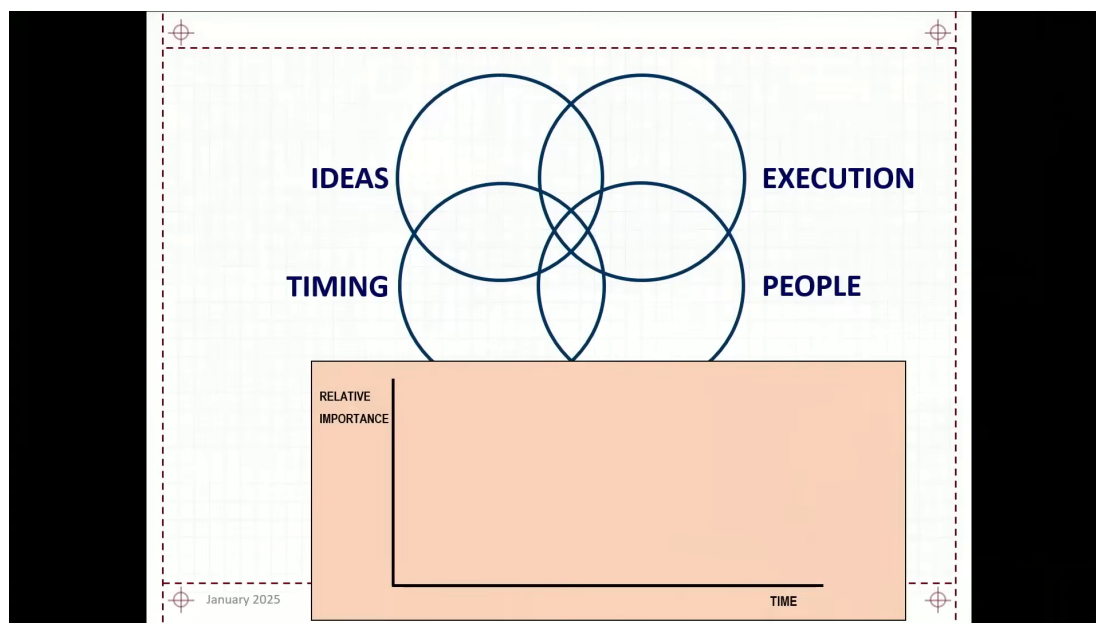


Figure 1.2: Four startup factors above an empty importance-time graph.

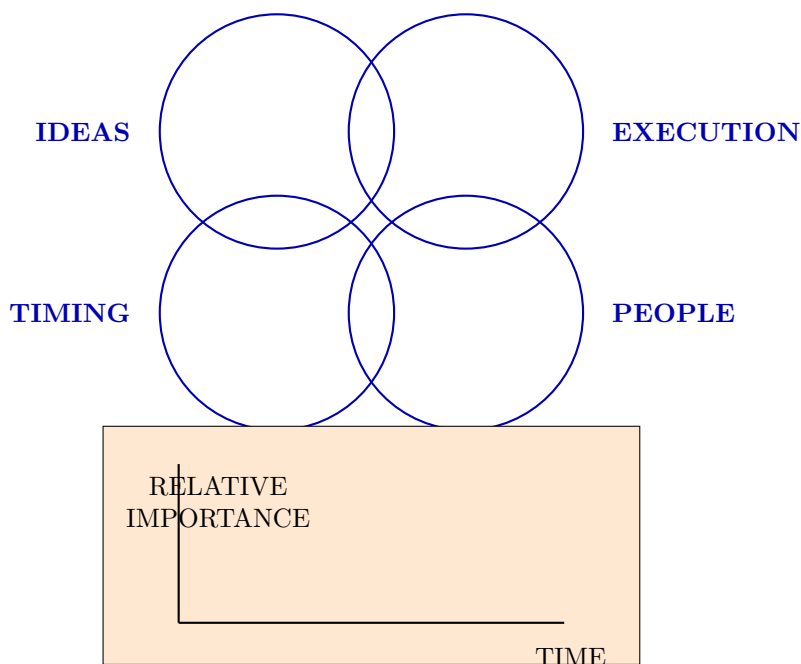


Figure 1.3: Pocket redraw of the four-factor framework before the lower graph is interpreted.

The upper panel says simultaneous alignment. The lower panel says, in effect, not yet. Although the lecture verbally introduces it as a “curve graph,” the visible slide at this stage is only a scaffold: axes, labels, and empty space. That is exactly right narratively. The lower panel is being reserved for a specific problem that has not yet been solved.

1.6.1 Ideas

The lecture is impatient with the mere existence of ideas. Around MIT, ideas are everywhere. The real question is whether an idea is valuable, to whom it is valuable, how much value it creates, and whether that value can be defended long enough to support a venture. An idea that can be copied instantly may still be admirable; it may even be socially useful. But the lecture insists that admiration and venture viability are not the same thing.

1.6.2 Execution

Execution is where the lecture becomes particularly concrete. Hadzima uses examples of deals he passed on not to boast about discernment, but to show how execution uncertainty feels before an outcome is known.

Zipcar looked conceptually important. Renting a car by the hour is a genuine break in the consumption model of transportation. But the operational burden, especially around parking and city-by-city implementation, was for him obscure enough to justify passing. The precursor-to-eBay story works the same way. The market concept was visible early, but the trust and fraud machinery required for large-scale execution was not yet persuasive.

This is why the lecture pauses over *mens et manus*. Ideas and execution belong together. Yet even that pair is not sufficient.

1.6.3 Timing

Timing is treated with unusual seriousness. Hadzima says plainly that he has lost more money and time by being ahead of the curve than in many other ways. Timing, in this lecture, is not a matter of trendiness. It is a matter of whether the surrounding world has become compatible with the venture.

3D printing is the clean example. The technology was not “new” in the simple sense. What had changed was the supporting environment: materials, sensors, microelectronics, and the expiration of earlier constraints. Prodigy, by contrast, spent heavily to do online shopping before the user environment and browser infrastructure were ready. Fusion makes the point in an even harder form. It does not work in pieces. All of it must work together, and that makes timing inseparable from the completeness of the system.

So when the lecture asks “Why now?”, it is not asking for a slogan about urgency. It is asking whether the world and the venture have arrived at the same time.

1.6.4 People

People, Hadzima says, are the single biggest source of failure in most ventures. He first frames this with the Cheshire Cat story: if we do not know where we are going, route choice is meaningless. He then makes the point vivid with the three-founder example. One founder wants the technology to become a standard, perhaps even open source. Another wants a fast-growing company that can go public. A third wants something important but more collegial and less intense. These are not three versions of the same ambition. They are three different journeys mistakenly housed in one venture.

The lecture then adds two more people problems. Sometimes teams are funded before they understand what they are doing, and then the pivot becomes internal rather than customer-facing. And sometimes teams are too certain too early. The E-Ink example belongs here. Not knowing what we do not know is itself a people problem because it determines whether a team is teachable.

1.6.5 Question & Answer

Question. How do technical and business contributions change over time inside a founding team?

Answer. The lecture answers this with a qualitative time-dependent graph. Early on, the technical side has very high relative importance because until something works technically, there is nothing to finance, market, or scale. The business founder, by contrast, may initially be working nights and weekends, perhaps still holding another job, trying to understand the market and the eventual commercial structure. Over time the balance changes. The business side rises in importance because distribution, team building, financing, negotiation, and organization become increasingly central.

The lecture's clean symbolic summary is qualitative rather than quantitative:

$$I_{\text{technical}}(t) \downarrow, \quad I_{\text{business}}(t) \uparrow. \quad (1.10)$$

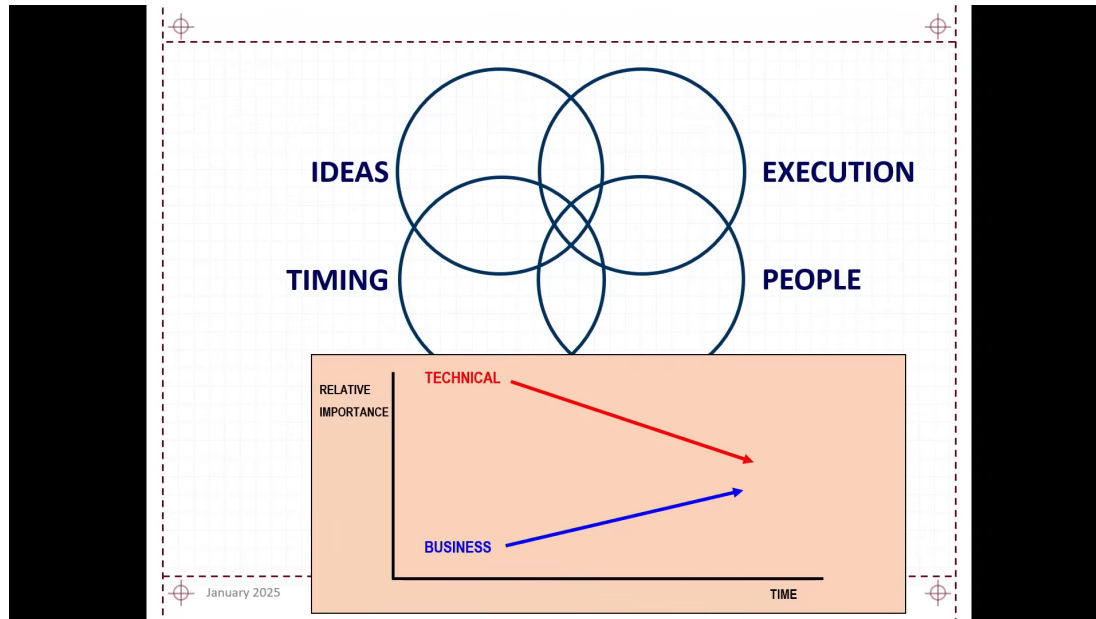


Figure 1.4: Technical importance falls as business importance rises.

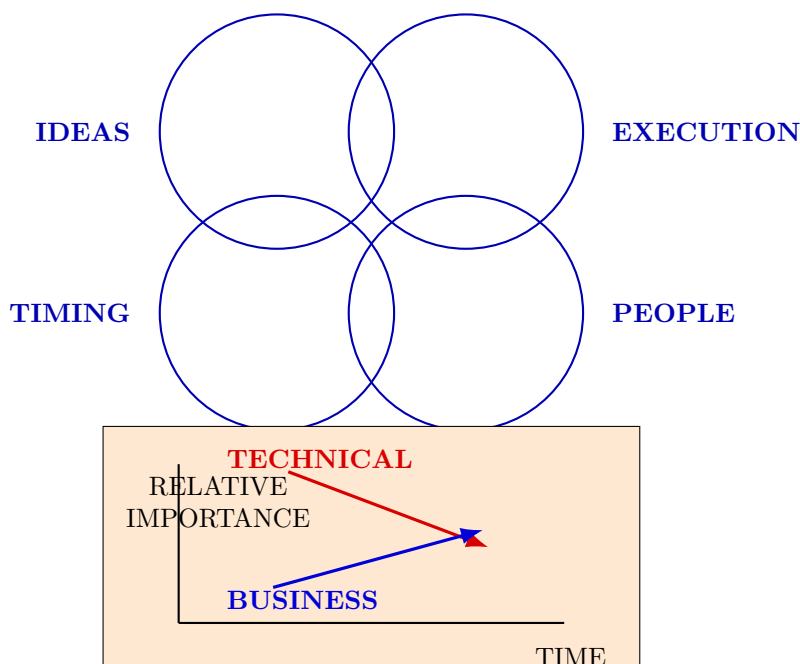


Figure 1.5: Pocket redraw of the lecture’s qualitative team-balance graph.

Remark 1.3. The graph is schematic. The arrows are straight, unscaled, and not tied to a marked crossing time. The lecture explicitly says that whether the lines cross is not the point. The point is that the relative balance changes, and with it the conversation about legitimacy, structure, and equity.

Once the graph has opened the team problem, the lecture does not leave the point in diagrammatic form. It immediately turns to actual ventures and asks which of the four components aligned and which failed.

SpeechCo. This is the lecture’s success case of alignment. The idea was strong. The timing was good because compute power had finally arrived. Execution improved by restricting the system to a domain of language use rather than trying to solve everything at once. And the people had spent enough time together before launch to form a functioning team.

VideoCo. This case initially looks like an execution failure in the making. The founders seem incapable even of choosing a company name. The payoff is that they were not merely naming the company; they were choosing a strategy. The DIVA/Avid connection reveals that what looked like hesitation had a directional logic.

HIV Co. Here the idea, timing, and scientific credibility were all strong. The weakness was people. Cofounder conflict drained the venture even in the presence of first-rate backers and advisers. This is Hadzima’s cleanest case of a venture weakened primarily by internal human structure.

NanoCo. This is the lecture’s timing case. NanoFuel had technical promise and environmental appeal, but the market arrived in the wrong price regime. The lecture even gives a back-of-the-

envelope threshold:

$$p_{\text{diesel}} \approx \$0.50/\text{gal}, \quad (1.11)$$

$$p_{\text{neutral}} \approx \$2.00/\text{gal}. \quad (1.12)$$

At launch,

$$p_{\text{diesel}} < p_{\text{neutral}}, \quad (1.13)$$

so the product fell below the lecturer's rough economic-neutral point. The lecture makes the reasoning even more vivid by observing that one could not buy a gallon of distilled water for \$0.50. The argument is intentionally simple, but it is enough. A technically interesting product may fail because the world supplies the wrong price at the wrong time.

1.7 From Vision to Pitch, Without Losing the Substance

Only after all this does the lecture turn to communication. The order is essential. We are not being taught to substitute rhetoric for venture substance. We are being taught how substance is compressed when it must be explained to other people.

Hadzima begins with the mission statement or vision, sometimes also treated as a value proposition. He then immediately insists that the vision must be supported. The lecture's pyramid is a hierarchy of compression.

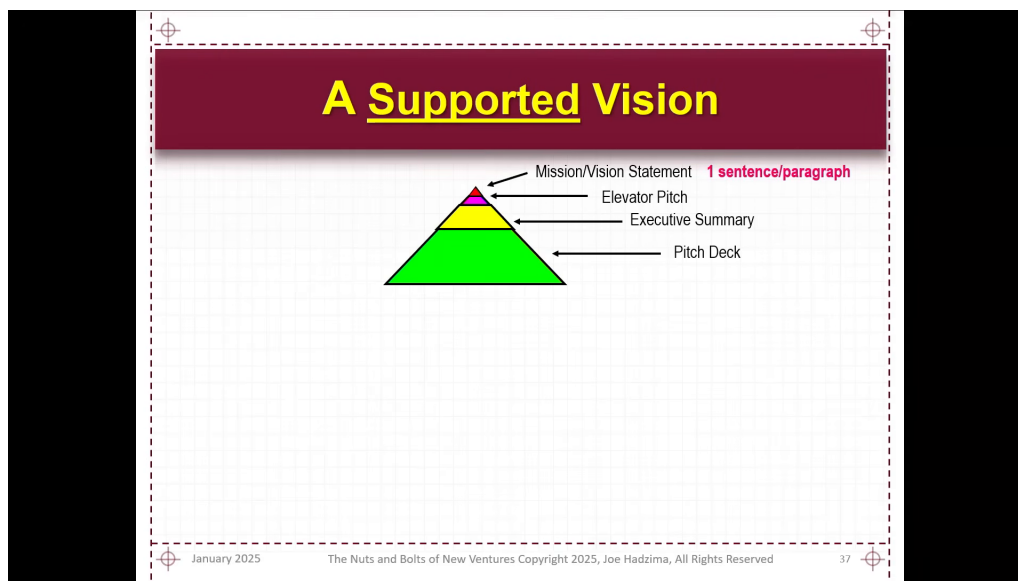


Figure 1.6: A supported vision from mission statement to pitch deck.

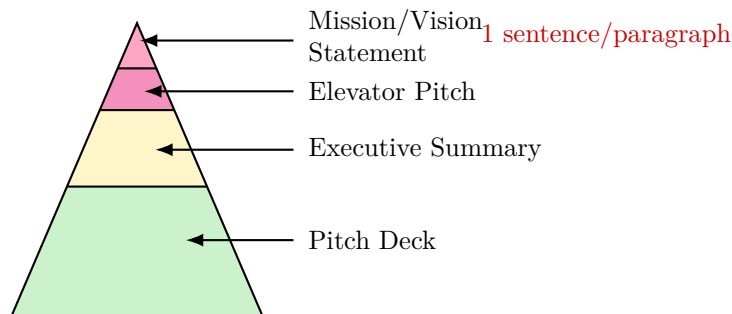


Figure 1.7: Pocket redraw of the supported-vision pyramid.

The original slide is worth keeping because the arrow targets fix the order of the tiers. The apex is the most compressed claim, and the lower tiers are progressively larger supports. One can read the entire device as repeated compression of the same venture at different bandwidths:

$$\text{mission statement} \sim 1 \text{ sentence or paragraph}, \quad (1.14)$$

$$\text{elevator pitch} \sim 30 \text{ seconds}, \quad (1.15)$$

$$\text{executive summary} \sim 1\text{--}4 \text{ pages}, \quad (1.16)$$

$$(\text{slides, minutes, font size}) = (10, 20, 30). \quad (1.17)$$

That last line is Guy Kawasaki's 10/20/30 rule: ten slides, twenty minutes, thirty-point font. Hadzima presents it as discipline. Each level asks the same question at a different scale: can the venture still make sense when compressed?

This is why the lecture then moves directly to the Theranos warning. Communication matters, but unsupported communication is not persuasion. It is deferred collapse. The lecture's earlier happiness equation becomes useful again. If reality is fixed while expectations are inflated, the ratio worsens:

$$R \text{ fixed}, \quad E_2 > E_1 \quad \implies \quad \frac{R}{E_2} < \frac{R}{E_1}. \quad (1.18)$$

So the lecture's final position is not anti-pitch. It is anti-separation. Sizzle without substance is empty, but substance that cannot be articulated will often fail to recruit the people, capital, and trust required for the venture to exist.

1.8 Summary

We can now see the lecture unfolding in the order Hadzima intended. It begins with motives, expands into operational ignorance, then creates the central tension by insisting on bad odds and painful realities. Only after that tension is fully present does it justify the course as a practical instrument for preparation. The room itself then becomes the site of the first equation, $H = \frac{R}{E}$, and that local rule about reality and expectations becomes a general rule about promises, delivery, and hype.

From there the lecture moves by successive compressions. First: create value and capture value. Then: why this, why now, why this team, and why won't this work? Then: ideas, execution, timing, and people. Finally: mission statement, elevator pitch, executive summary, and pitch deck. Each

compression is followed by examples, because the lecture never allows a framework to float free of lived ventures.

The ending is modest, and that modesty is part of the seriousness of the lecture. Luck remains. Preparation does not abolish it. It changes the odds of being ready when luck matters. That is why the lecture ends not with startup triumphalism but with a practical handoff: break the room into interest groups, meet collaborators, return for logistics, and then go find customers.

Chapter 2

Market Identification and Sales: Finding Your Customer

These notes follow Lecture 2 of MIT 15.393, in which Joseph Hadzima handles the course machinery and then turns the room over to Bob Jones for a sustained discussion of finding customers. The curation is by LazyingArt LLC. No validated mathematical frames survive from this lecture, so every displayed relation and every figure below is a cautious reconstruction from the transcript. The lecture's method is itself part of the lesson: begin with a case so simple that we are tempted to dismiss it, and then keep asking the next honest question until the real business either comes into focus or falls apart.

2.1 Logistics, Teams, and the Hand-Off to Customer-Finding

The lecture opens with logistics, but the logistics are already saying something entrepreneurial. Hadzima speaks about attendance, the written requirement, and the preference that the work be done in teams. One reason is comic and administrative: fewer separate submissions are easier to review. The serious reason is the one the lecture wants us to keep. Teams, he says, have a higher probability of success than sole entrepreneurs, which we may summarize as

$$\Pr(\text{success} \mid \text{team}) > \Pr(\text{success} \mid \text{solo}). \quad (2.1)$$

That inequality is a compact formalization of a spoken claim, not a theorem proved in the room; but it is the first practical relation of the evening, and it explains why the lecture begins with team formation rather than with product genius.

Only after the course structure is in place does Hadzima hand the room to Bob Jones. Jones immediately makes a contract with the audience. If he is going to ask for ninety minutes of attention, he says, he ought to return something worthwhile for that investment. He therefore asks two questions before he teaches anything substantive: who are you, and what do you want? The room turns out to be mixed. Some people are already on a second venture, some are in their first, some are seriously considering one after graduation, and some are simply exploring. The jokes matter here: medication is available for first-time founders; exploratory work can be as unacademic and as useful as springboard diving during IAP. The room is heterogeneous, and Jones wants the argument to be usable across that range.

This is why he makes his anti-academic turn. He mocks the sort of lecture that begins with a sentence about risk aversion, mean-variance utility maximization, and quadratic utility functions, gets dutifully written down, and leaves nobody any wiser. That joke is not the content of the lecture. It is the permission for the method. Jones tells us he will go in the opposite direction: start with examples so simple they are almost trivial, and let the complexity arrive only when the example itself demands it. That is the pedagogical pivot on which the whole chapter turns.

2.2 Start at the End: Money Needed, Customer Value, Customer Count

Jones now gives us the toy model. Suppose, he says, that he flees the overcharged intellectual world, moves to the Berkshires, becomes a grumpy old man in a tiny house, and decides to give guitar lessons. He can imagine the whole thing very easily: beginner chords, tuning, groove, rhythm, more advanced work, even how to play a mixolydian solo that still sounds like the blues. He would need a small studio, a few amplifiers, perhaps a drum machine. Then he stops himself: let us stop right there.

2.2.1 Question & Answer

What mistake did he make? The answer is the first conceptual obstacle of the lecture. We are often quick to ask whether we *could* do a venture. Jones insists that we must first ask whether we *should* do it. The bridge from the first question to the second is money. Most companies die, he says, because they run out of money or because they are on a path where they will never make any. So he reverses direction and begins at the end of the story. We want to make enough money. That means we need revenues, and those revenues should exceed our costs:

$$R > C. \quad (2.2)$$

Revenues come from customers. So the real problem is not “I have a product idea” but “how many customers do I need, and is that number sane?”

The notation here is a cleaned reconstruction of Jones’s spoken arithmetic, not notation visibly written in the room. Let M denote the amount of money we need over a chosen period, and let V_c denote the value of one customer over that same period. Then the required customer count is

$$N = \frac{M}{V_c}. \quad (2.3)$$

In this lecture V_c is not a grand lifetime-value formula. It is simply what one active customer is worth on the same monthly or yearly clock used for M .

Jones gives the numerical skeleton in one line:

$$\frac{1,000,000}{1,000} = 1,000. \quad (2.4)$$

If we need a million dollars and each customer is worth a thousand, then we need a thousand customers. Only after that division do we ask the practical question: how are we going to do that?

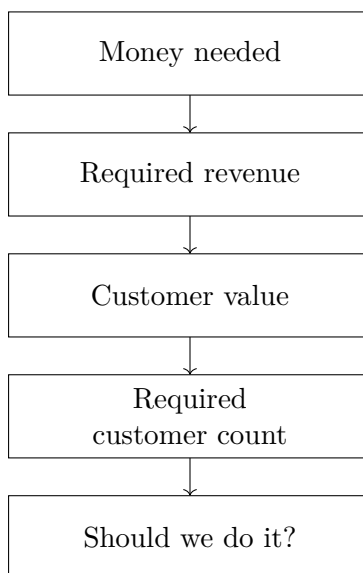


Figure 2.1: Transcript-derived reconstruction of Jones’s backward logic. We do not start from enthusiasm for the offering; we start from the money requirement and work backward until the venture either looks feasible or does not.

Now the toy business is run through the arithmetic. Jones wants only a modest living. On a monthly time base,

$$M = \$5,000/\text{month}, \quad (2.5)$$

$$V_c = 25 \text{ \$/hour} \times 2 \text{ sessions/month} = 50 \text{ \$/customer/month}, \quad (2.6)$$

$$N = \frac{M}{V_c} = \frac{5,000}{50} = 100. \quad (2.7)$$

A hundred customers is not frightening for a large organization. It is a very different number for one grumpy guitarist starting alone. Jones does not hide that discouraging answer. The arithmetic is supposed to hurt a little. That pain is the point. It is cheaper to discover the problem in five lines of arithmetic than in two years of effort and most of a life savings.

2.3 Users, Payers, Champions, and What We Are Really Selling

Having forced the first go/no-go calculation, Jones now asks the question he had postponed: what exactly is broken here, and who pays us to fix it? The default answer is schoolchildren. They are the obvious users. But they are not the obvious buyers. Past a certain point, Jones says quite plainly, the users are not the customers. The parents are the ones who pay. Once we notice that, the economics remain the same but the meaning changes. The child may want fun, status, escape from boredom, or maybe music itself. The parent may be buying activity, discipline, social development, or merely an experiment.

The lecture then performs its first major reset. Jones says, in effect, erase that and start over. What if the relevant customer is not the child but the adult who used to play, used to sing, used to feel cool, and misses all of that? He gives a much richer version of the service: teach a recognizable tune on Tuesday, assemble an ensemble, and let the person play it with others on Thursday. The

Customer definition	Primary payer	Approx. customer value per month	Customers needed for \$5,000/month
Schoolchildren taking lessons	Parents	\$50	100
Lapsed adults wanting to play again	Adult learner	$\approx \$400$	≈ 12.5
Older adults seeking music and company	Adult learner	$\approx \$400$	about a dozen

Table 2.1: Transcript-derived comparison of three customer definitions for roughly the same underlying teaching activity. The lesson is that the economics change when the payer and the perceived benefit change.

ticket is no longer mere instruction. It is competence regained, identity recovered, and performance made possible. The arithmetic shifts with it:

$$V_c \approx 50 \text{ \$/hour} \times 2 \text{ sessions/week} \times 4 \text{ weeks/month} \quad (2.8)$$

$$\approx 400 \text{ \$/customer/month}, \quad (2.9)$$

$$N \approx \frac{5,000}{400} = 12.5. \quad (2.10)$$

Jones rounds the answer in words: about a dozen or so.

He pushes the point once more with the older adults he has seen through volunteer music work. Here the same visible format meets yet another need: social life, purpose, companionship, getting out of the house, perhaps even the possibility of romance. The lesson is not that one should randomly mutate the market. It is that changing the customer definition changes the economics because it changes the benefit being purchased.

2.3.1 Question & Answer

Who wins if I win? Jones slows the room down over this question because it widens the frame. The first answer is customers, and that answer is correct but incomplete. Customers may become a sales force by talking to one another. End users may become champions. Economic buyers may not be the same people as the users. Adjacent businesses may benefit from our success.

His music-store example is the cleanest illustration. A lapsed guitarist goes into a shop to buy strings, stares at the instruments on the wall, and explains that he has lost his chops and has no real setting in which to play. The helpful person in the store says, in effect, I know a guy who can fix that. If the customer starts taking lessons, the store may later sell strings, better instruments, and more equipment. We win; the store wins; the player wins. This is not generic referral marketing. It is a structural point about aligned incentives.

At this stage Jones asks the other crucial question: what are we really selling? One student says romance. Jones half laughs, half accepts, and then clarifies. We are selling comfort, socialization, purpose, collaboration, and the excitement that comes when ensemble performance suddenly clicks. The anecdote about Jones himself walking into a blues jam, playing with strangers, and feeling “magic” happen is not decoration. It is a phenomenology of the benefit. The guitar lesson is the visible product. The invisible product is the life the customer gets to step back into.

2.4 The Non-Negotiable Requirement: Customers

Having extracted a surprising amount from a tiny guitar business, Jones tightens the screws. He now lists the venture questions in a blunter form. We should ask:

1. What is broken?
2. Who specifically wants a solution?
3. Are there enough of those people?
4. Will they spend money to solve the problem?
5. How are they solving it now?
6. Why is our answer compellingly better than the alternatives?
7. Who actually pays?
8. How do we find them and let them know about us?

The questions are easy, he says. The answers are not. But if we do not at least face these questions, we will drive straight into the hole and only discover it when it is too late.

This is also the moment where the lecture pivots toward investor logic. Jones repeats Hadzima's hard framing: most investors lose most of their money. He jokes that perhaps only eight out of ten fail at MIT instead of nine out of ten elsewhere, but the odds are still dreadful. So the investor wants the answer to two questions: how will you make money, and how will I make money? The order matters. If the venture cannot answer the first question, there is no point even asking the second.

From that perspective, there is only one non-negotiable requirement. We must have customers. Everything else can be renegotiated. Jones says this with the force of a room trying to get away with something and being told that it cannot. We do not have to be rich. We do not have to be beautiful. We do not even have to look especially brilliant. If we have customers, the rest of the world begins to reinterpret us. That is the bridge into the first serious case study. Enough of the abstract, he says. Here is what it looks like when smart people get this wrong.

2.5 Customer Discovery by Failure: The Regain Case

The first medical case arrives only after the scaffold is in place, and the sequencing matters. Jones wants us to hear how reasonable a bad plan can sound. He was running the clinical nutrition division of a medical company in California, a division whose economics were excellent:

$$\text{cost per liter} \approx \$3, \quad \text{sale price per liter} \approx \$70. \quad (2.11)$$

The broader company had been doing roughly two hundred million dollars a year for a long time, but managed care threatened the old profit pool. So Jones went looking for the next revenue source and found, as he says, an opportunity nobody else was pursuing. He immediately adds the correct diagnosis in hindsight: that should have been a red flag.

The field was kidney dialysis. The background numbers looked attractive:

$$400,000 \text{ patients}, \quad 2,300 \text{ dialysis centers}, \quad 8\text{--}12 \text{ lb fluid gain between treatments}. \quad (2.12)$$

The contradiction also looked obvious. Dialysis patients often need nutritional supplementation, but the standard supplements were liquids, precisely the thing these patients should not be loading up on. So the company designed a bar-like product, high in what patients needed, low in what they should avoid, and without added fluid. The product was named *Regain*.

The company did the respectable things. It ran a clinical trial even though it was not required. It published the result. The result was good: blood chemistries improved, lives were extended. It then asked opinion leaders, clinicians, and dieticians whether they would recommend the product. The answer, extraordinarily, was yes. For how many patients? All of them. How often? Seven days a week. At what price? Roughly \$3 per bar. The arithmetic looked almost too good. Jones says they began to think they had replaced the business they were afraid of losing.

Then the product was launched, and the floor dropped out:

$$\text{\$400,000 forecast} \quad \text{versus} \quad \text{\$35,000 actual.} \quad (2.13)$$

As a cleaned-up ratio,

$$\frac{35,000}{400,000} = 0.0875 < 0.10. \quad (2.14)$$

Month after month, the product came in at less than ten percent of forecast.

2.5.1 Question & Answer

What did we do wrong? Jones lets the question sit in the room because the answer comes in layers.

1. We interviewed clinicians, but the patients were the buyers.
2. We judged taste with the wrong palate.
3. We mistook clinical need for willingness to pay.
4. We misunderstood the motivational composition of the market.
5. We forgot that retailers were customers too.

The first two mistakes are concrete. Clinicians did not buy the product and did not eat it. Patients did both. When Jones's team finally spoke to patients, they learned that many did not want the product and could not afford it. The taste problem was equally sobering. In kidney failure the palate shifts, and a heavily protein-based product can taste like spoiled meat. The team had asked the wrong people how it tasted.

The third and fourth mistakes cut deeper. Jones says the real question should have been: who gets kidney failure? The transcript gives a rough decomposition,

$$33\% + 44\% = 77\% \approx \frac{3}{4}, \quad (2.15)$$

which he interprets as follows: about three quarters of the customer base had arrived there through long-term mismanagement of diabetes or high blood pressure. His conclusion is brutal but plain. A great many of these patients had not taken care of themselves before. Many were not going to start now. Clinical benefit was real, but motivation to spend was weak.

The final mistake is the one the lecture keeps widening toward. Patients still had to buy the product somewhere. Jones's company understood hospitals; it did not understand retail pharmacy.

That ignorance was expensive. A pharmacy manager has only seconds to hear a pitch. A shelf is a monetized asset. Remove a known product, insert an unknown one, and the retailer wants compensation. That is the logic of slotting fees, and Jones gives the number that made the lesson unforgettable:

$$\text{\$1,000,000 per quarter for CVS,} \quad (2.16)$$

before guaranteed sales, refunds, and chain-wide stocking requirements were even fully counted.

The failure therefore was not one error but a stack of errors, each caused by a different bad definition of customer. Jones later found a way to reach the roughly twenty-five percent of the market that had not arrived through self-neglect, and the broader company recovered. But the lesson of the year is not rescue. It is diagnosis. The consequences of not thinking through the customer system got worse and worse.

2.6 Rebuilding the Business Around the Real Customer: Night-time Diabetes

With characteristic dark humor, Jones says that entrepreneurship may be a form of mental illness, because after all this he went off and did more of it. The next medically grounded venture is the constructive answer to the Regain failure. This time the company enters diabetes, and the starting point is not a product gap but a lived problem.

The market is large:

$$10,000,000 \text{ diagnosed patients,} \quad 4,000,000 \text{ insulin users.} \quad (2.17)$$

The practical aim of diabetes management, as Jones states it, is to keep blood glucose from fluctuating wildly. If we denote the overnight glucose level only schematically by $g(t)$, then we should read the next picture as a qualitative shape, not as a clinical graph with calibrated axes. During the day, disciplined patients can spread intake and insulin across small meals and small snacks. At night that discipline is much harder to sustain. Instead people tend to eat a large bedtime snack, take a large insulin bolus, and hope the two will titrate through the night.

The hope fails for a simple reason. The food turns to glucose quickly. The spike is higher, but the support does not last longer. Jones identifies the dangerous interval as

$$2 \text{ a.m.} \leq t \leq 6 \text{ a.m.}, \quad (2.18)$$

the period in which food support has run out but insulin may still be working.

At this point the lecture changes pace. Jones explains focus groups, then rebels against the standard focus-group script. After too much science and too much intellectual talk, he bursts out and asks what the customers are actually afraid of. The answer is the first true emotional discovery of the case: some people no longer sleep in a bed because they are afraid they will get too comfortable and die in their sleep. That is the sentence around which the business is rebuilt.

So the company creates a product whose components convert to glucose at different rates and last all night. Jones says that, being marketers at heart, they called it time-release glucose. Later in the lecture he refers to the resulting business by product name, *Night Bite*. This time, he adds pointedly, it also tasted good.

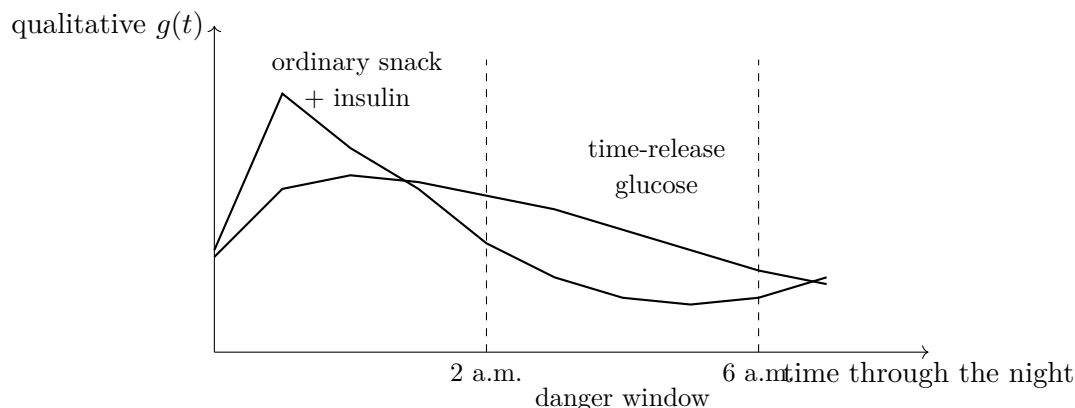


Figure 2.2: Transcript-derived schematic of the overnight problem. The ordinary bedtime snack gives a sharper early spike without widening the safe interval, whereas the redesigned product is meant to spread glucose availability further into the night.

2.6.1 Question & Answer

Why would putting “for diabetes” on the label kill the business? Jones raises this as a genuine puzzle because he originally fought to be able to say exactly that on the package. The customers corrected him. It is nobody’s business, they say, that I have diabetes. I am not “a diabetic” in the public sense; I am a banker, a lawyer, an entrepreneur, an active professional. If I pull out something that looks like medicine during a meeting, people will think I am sick, frail, and professionally diminished. That is a career-limiting signal.

So the product has to look like something an elite athlete might use, not like medication. The chemistry can be medical; the public identity cannot. That is why the company gives it a non-medical name and why Jones later notes, with some satisfaction, that marathon runners ended up buying it too.

The redesign also becomes more precise because the company now talks not only to clinicians but to parents and patients. The unit is made deliberately small and divisible:

$$100 \text{ calories per unit,} \quad (2.19)$$

with scored use that allows 50, 100, or 150 calories depending on the user. A better product has now emerged. But Jones is careful not to stop there. A better product is still only half a business. The next problem is distribution.

2.7 Channels, Free Sales Forces, and Selling Customers to Retailers

Jones now slows the story down and turns to what he calls the nuts and bolts. The first discovery is that channel changes perceived value. In a focus group the company puts paper in front of users and asks two questions: where would you expect to buy a product like this, and how much would you expect to pay? The answers sort into two piles:

$$\$0.49 \text{ in a grocery store,} \quad \$1.25 \text{ in a pharmacy.} \quad (2.20)$$

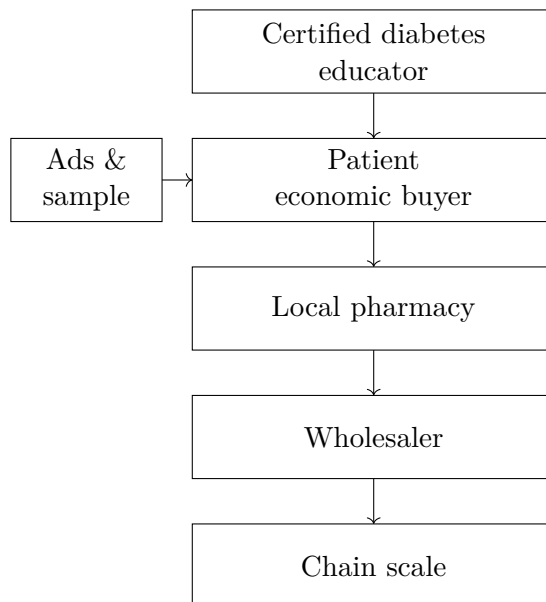


Figure 2.3: Transcript-derived reconstruction of the Night Bite customer system. The patient is the economic buyer, but the educator, the pharmacy, and the wholesaler each carry distinct incentives. The venture only scales when those roles are aligned rather than collapsed into one vague “customer.”

Same product, different channel, different value. Jones says, dryly, call me silly, I like pharmacies.

The next discovery is organizational. A business like this does not sell to “the market” in one undifferentiated move. It sells through a customer system. The crucial intermediary turns out to be the certified diabetes educator. These professionals can literally be found by zip code through their association. More importantly, they have a professional frustration: patient compliance. They tell patients what to do, and the patients do not do it. If Jones can give them something patients will actually use and thank them for recommending, he has found people who win if he wins.

Several thousand educators eventually become a kind of unpaid sales force. Jones is careful about why. They are not unpaid because they are fools; they are unpaid because they are healers, and what they want is not commission but something their patients will comply with. The educators can then recommend the product to patients and even help pressure local pharmacies to stock it.

The company first sells directly, not because that is the intended endpoint but because Jones wants information. He wants reorder frequency so he can infer usage. Advertising generates inbound calls, and the numbers become a kind of live demand function:

$$100, 200, 300, 500 \text{ calls/day.} \quad (2.21)$$

At the same time, long-run usage turns out to be strikingly strong:

$$\text{reuse rate} > 300 \text{ occasions/year.} \quad (2.22)$$

That is not a market-research fantasy. It is behavioral evidence.

Jones also gives a small methodological lesson from inside sales. Closed-ended questions stop the conversation too quickly. “Did you like it?” can end with “no.” An open-ended question such as

“What did you think?” keeps the door open long enough to discover that taste may have been imperfect, efficacy was good, and the customer wants to reorder anyway.

The pharmacy-entry tactic is equally concrete. A store manager may have a purchase amount below which corporate approval is unnecessary. Jones guesses that threshold is about one hundred dollars and prices the starter kit just below it:

$$\$99.95 < \$100. \quad (2.23)$$

The risk is removed: if it does not sell, throw it away and get a refund. Then the company calls its existing customers and tells them where the product is now stocked. When the first batch sells out, the pharmacy itself becomes interested. The phrase Jones uses is exact and worth preserving: he bought a bunch of customers and then sold those customers to the retailers.

The same logic is pushed upstream. A large pharmacy may carry about eight thousand SKUs and does not want eight thousand invoices. So the company gets into wholesalers by seeding nearby pharmacies, giving the wholesaler its normal margin, and asking only to be entered into the system before reorder demand appears. Once that is done, scale becomes possible.

The planogram meeting at a major chain is the final demonstration. Jones and his colleague go into a fifteen-minute meeting with no money to pay slotting fees. They spend ten minutes saying who they are and why the product is good, and then open a briefcase full of printouts showing ten thousand existing diabetes customers in the states where the chain operates. Those customers all have one question: which pharmacy carries the product? How, they ask the buyer, would you like us to answer them? Jones then says nothing. He invokes the negotiating rule of thumb that the next person who speaks loses. After the silence, the buyer says yes. What won the meeting was not persuasion in the abstract. It was customers.

2.8 Segment by Motivation, Not Demographics

By the end of the Night Bite story the lecture has earned the right to become doctrinal. Jones turns back to the room and asks what he should have learned from all these experiences.

2.8.1 Question & Answer

The audience answers, and Jones sharpens the answers as he goes. The resulting principles are the real retrospective of the lecture:

1. Talk to your customers.
2. Let the customer definition evolve as the business evolves.
3. Reinvest early effort and money in better market knowledge.
4. Treat distribution as part of the venture, not as an afterthought.
5. Figure out what people are really buying.

That last point becomes a short theory of benefits. Nobody specifically wants the product. People want what the product does. Jones’s example is vision: we may hire glasses, contact lenses, or a LASIK surgeon, but what we really want is to see better. The same principle reaches backward

through the entire lecture. Nobody wanted “guitar lessons” in the narrow sense. They wanted social life, competence, romance, purpose, performance, collaboration. Nobody wanted “glucose” in the narrow sense. They wanted safety, sleep, and the ability to function without public stigma.

Jones then asks what market segmentation really means. Conventional marketers segment by geography, income, or gender. He argues that the entrepreneur, especially the undercapitalized entrepreneur, must segment by motivation. The transcript’s cleanest example is the product that prevents a second heart attack. The person who has just survived the first one sits at the top of the motivation pile. Nearby family members may listen next. Social peers may pay some attention after that. The broad public is the low-motivation block at the bottom:

$$\text{most motivated} > \text{related observers} > \text{adjacent social group} > \text{general public.} \quad (2.24)$$

That ordering is qualitative, not measured. Its meaning is strategic. A startup cannot afford to spend scarce resources trying to convert the least motivated part of the world. Jones calls that big bottom block a graveyard.

The comparison between Regain and Night Bite makes the principle vivid. Regain improved clinical markers, but the patient felt no immediate difference. Every six months the doctor might say, nice going, your blood chemistries are better. Night Bite produced immediate morning feedback: I think it worked. In Jones’s joking form, the user did not wake up dead. The line is facetious, but the point is exact. Motivation rises when the benefit is not only real but felt.

He then compresses his own view of marketing almost to absurdity. After acknowledging all the sophisticated mechanics, he says he can save us two years of graduate school with two bullet points, and the two bullet points are the same: find out what your customers want; then find out what your customers want. The repetition is not carelessness. It is emphasis.

The very last quick story before the closing homework lands the lesson one final time. Jones says a group of them once invented a sleep-support product and, in a foolish top-down move, decided the market was working moms. They assembled a huge list. They heard nothing. Then a few repeat purchasers emerged, and when Jones called them personally and asked why they were buying, every answer turned out to be some version of the same thing: I run marathons, I need better recovery, better sleep helps me recover, and therefore your product helps me perform. Jones thought he was selling sensitivity and care. He was selling to warriors. He says flatly that he would have died of old age before learning that from a sequence of Google ads. Talk to them.

That is why the lecture ends by narrowing, not widening. The next session will be about pitching. The homework is to pitch the business in three minutes, then in thirty seconds, then, if possible, in three sentences and about ten seconds. The compression is not a new topic. It is the compressed form of everything in this lecture.

2.9 Summary

The lecture begins with attendance codes and team formation, but its real structure is mathematical and strategic. We begin at the end of the story, ask how much money we need, estimate what one customer is worth, and derive how many customers are required. Then we ask the harder questions: who is the user, who is the payer, who is the champion, who else wins if we win, and what benefit is actually being purchased?

The Regain case shows how completely a business can fail even when the clinical result is real

and the arithmetic looks encouraging. The Night Bite case shows how the same kind of medically grounded business can succeed once fear, identity, channel, and reorder behavior are taken seriously. The final principle is severe and practical: do not aim at everybody. Find the people who want the offering most, learn what they are really buying, and build the venture around them.